



LXD: The Pure Container Hypervisor

A hypervisor is computer software, firmware or hardware that can create virtual machines. LXD is bundled with Ubuntu 16.04, and has the advantage of being able to run hundreds of unmodified Linux operating systems on a single server and at incredible speeds.



When Ubuntu 16.04 was launched, one of its most interesting features was software called LXD that was included with it. LXD is somewhat similar to Docker but differs from it in a few aspects.

The base technology used in both Docker and LXD is LXC or Linux Containers, which has been around for a few years. It didn't witness much traction because of the complexity involved in setting up the containers. Both LXD and Docker provide a high level user-friendly interface to the low level commands offered by LXC.

Differences between LXD, Docker and KVM

The primary difference between LXD and Docker is that the latter is meant for application-specific containers. Consider a typical use case of a Web application. You'll need a database server and a Web server. So you will have two Docker containers—one which runs the database and the other running the Web server. The containers do not perform any other function.

In LXD, it is possible to run a complete OS as a container. For this, typically, one requires KVM or Xen and full hardware virtualisation (emulated disk, network cards and CPU). Due to the number of layers in full HVM, the performance is not as good as LXC but security is better. The guest virtual machine is fully isolated from the host machine,

which decreases the attack surface. Ubuntu officially supports running RHEL, CentOS, Ubuntu, Debian and Oracle Linux containers in LXD.

Setting up LXD

LXD is best used with the Z File System (ZFS), which has become relatively stable. Ubuntu 16.04 supports ZFS officially. ZFS is not supported as a root file system in an Ubuntu installation—you don't get that option even while installing Ubuntu, but you can always use it as a data storage file system.

The advantage of ZFS is that you get quick CoW (Copy-on-Write) snapshots, clones which are very useful for containers. Let's assume that you have created a container and installed the LAMP stack. Now, you can snapshot it and create as many LAMP containers at no time just by cloning them. The best part is that the clones consume disk space only for what differs from the original data. If you have installed your WordPress application in one of the containers and, let's say, WordPress takes up 10MB, then that particular container will consume only 10MB, unlike a full HVM for which you have to allocate a complete 10-20GB disk image file.

Creating the ZFS pool

Assuming you have a fresh installation of Ubuntu, with a spare disk or partition in the existing disk, we can install the required